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Dual-Mode Pressure-Sensing and Illumination Mat

Abstract

Systems and apparatuses for dual-mode pressure-sensing and illumination mats are provided. The systems include a surface that includes a plurality of row wires and a plurality of column wires forming intersection points. At the intersection points, the systems include a pressure sensor configured to sense a pressure exerted upon the surface and a light emitting diode (LED) connected in parallel. The systems include a controller coupled to the selector and configured to control operation of the surface, wherein the controller is configured to operate in one of a pressure-sensing mode or an illumination mode. When the controller is configured to operate in the pressure-sensing mode, the controller scans pressure values provided by the pressure sensors. When the controller is configured to operate in the illumination mode, the controller configures the surface to enable the LEDs in accordance with an illumination pattern.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a national stage of Patent Cooperation Treaty Application No. PCT/US2023/34677, filed on Oct. 6, 2023 which claims priority to U.S. Pat. No. 11,895,754, filed on Nov. 30, 2022, and which are incorporated herein by reference in their entirety.

BACKGROUND

[0002] Frictionless checkout systems typically rely on cameras positioned throughout the venue to detect which items to associate with shoppers. For example, if a shopper picks a product off a shelf, the frictionless checkout system may add the product to an inventory of items associated with the shopper. As a result, when the frictionless checkout system detects that the shopper has left the venue, the system may automatically process the purchase of any items in the shopper's inventory.

[0003] To detect when shoppers pick items off venue shelves, venue operators may deploy pressure mats configured to sense a change in pressure. These pressure mats conventionally include an array of pressure sensors to more precisely detect a location associated with the change in pressure. As a result, the frictionless checkout system may identify a refined area of interest within the image data generated by the cameras to analyze and identify the product picked up by the shopper and/or confirm that the shopper picked up an item assigned to the location by a planogram.

[0004] As part of configuring a frictionless checkout system, the cameras positioned throughout the venue need to be calibrated to ensure that the frictionless checkout system knows the correlation between the pixel location and venue location. Typically, the calibration process involves detecting illumination events at known locations of the venue.

[0005] However, conventional pressure mats do not include illumination assemblies. As a result, a venue operator configuring a conventional frictionless checkout system needs to install standalone illumination systems to calibrate the cameras. For example, the venue operator may install shelf-mounted lighting and/or under-shelf illumination systems. These solutions suffer several drawbacks. Notably, these standalone illumination solutions require their own power sources, mounting structures, and/or hardware assemblies that result in a higher cost of operation, longer configuration times, and/or less space available for products.

[0006] Accordingly, to overcome at least the aforementioned challenges, there is a need for a dual-mode pressure-sensing and illumination mat.

SUMMARY

[0007] In a first embodiment, the present invention is an apparatus comprising (1) a surface that includes a plurality of row wires and a plurality of column wires forming intersection points, wherein intersection points between row wires and column wires include (a) a pressure sensor (i) coupled to the row wire and the column wire at the intersection point and (ii) configured to sense a pressure exerted upon the surface, and (b) a light emitting diode (LED) coupled to the row wire and the column wire at the intersection point in parallel to the pressure sensor; (2) a selector configured to selectively complete an electrical path the flows from a power supply through a selected intersection point; (3) a controller coupled to the selector and configured to control operation of the surface, wherein the controller is configured to operate in one of a pressure-sensing mode or an illumination mode; (4) a switch bank coupled to the controller and the surface, wherein the controller is configured to operate the switch bank in tandem with the selector when operating in

the pressure-sensing mode; (5) a resistor bank coupled to the controller and the surface, wherein the controller is configured to operate the resistor bank in tandem with the selector when operating in the illumination mode; and (6) a switch operatively coupled to the controller, wherein (a) when the controller is configured to operate in the pressure-sensing mode, the controller operates the switch such that the switch completes a circuit between the selector, the surface, and the switch bank, and (b) when the controller is configured to operate in the illumination mode, the controller operates the switch such that switch completes a circuit between the selector, the surface, and the resistor bank. [0008] In another embodiment, the present invention is a modular apparatus comprising (1) a first apparatus configured in accordance with the first embodiment and configured to operate as a supervisor apparatus; and (2) a second apparatus configured in accordance with the second apparatus and configured to operate as a subordinate apparatus; wherein the controller of the first apparatus is coupled to the controller of the second apparatus, and the controller of the first apparatus is configured to (a) transmit illumination patterns to the second apparatus, and (b) receive indications of sensed pressure values from the second apparatus.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.

[0010] FIG. 1 illustrates a perspective view of an example venue that includes a product monitoring system, as disclosed herein.

[0011] FIG. 2 illustrates a perspective view of a shelf that includes a dual-mode mat being monitored by the product monitoring system, as disclosed herein.

[0012] FIG. 3A illustrates an example schematic diagram of a dual-mode mat, as disclosed herein.

[0013] FIG. 3B illustrates an example schematic diagram of a surface matrix included within a dual-mode mat, as disclosed herein.

[0014] FIG. 3C illustrates a side view of a dual-mode mat, as disclosed herein.

[0015] FIG. 4A illustrates an example view of a dual-mode mat being configured to illuminate in accordance with an illumination pattern, as disclosed herein.

[0016] FIG. 4B illustrates an example data structure for an illumination pattern, as disclosed herein.

[0017] FIG. 5A illustrates an example schematic diagram of an alternate embodiment of a dual-mode mat, as disclosed herein.

[0018] FIG. 5B illustrates an example diagram of a surface matrix included within the alternate embodiment of a dual-mode mat, as disclosed herein.

[0019] FIG. 6 is a block diagram representative of a controller of a dual-mode mat and the various components thereof.

[0020] FIG. 7 illustrates a modular arrangement of dual-mode mats coupled via a communication bus, in accordance with techniques described herein.

[0021] FIG. 8 illustrates a modular arrangement of dual-mode mats coupled via a daisy chain arrangement, in accordance with techniques described herein.

[0022] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

[0023] The apparatus and method components have been represented where appropriate by

conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

[0024] FIG. 1 is a perspective view, as seen from above, of a venue **100** illustrating an arrangement for which a product tracking system (e.g., a frictionless checkout system) within the venue **100** is deployed to facilitate tracking objects within the venue **100**. Although the example venue **100** is a retail venue, other types of venues (such as a warehouse, a transport center, an airport, a stadium, a museum, and so on) are envisioned. In the example embodiment of FIG. 1, the venue **100** includes a backroom that has a central controller **16**. In other embodiments, the central controller **16** may be located elsewhere on or proximate to the venue **100**. The central controller **16** may be operated by venue personnel and/or personnel associated with the RFID system.

[0025] The central controller **16** may comprise a networked host computer or server. The central controller **16** may be connected to a plurality of cameras **30**, **31** positioned throughout the venue **100** via the network switch **18**. For example, the plurality of cameras may include ceiling-mounted cameras **30** and wall-mounted cameras **31**. The plurality of cameras **30**, **31** may be positioned at a fixed location and/or support pan/tilt/roll control. Additionally, the venue **100** includes a plurality of dual-mode pressure-sensing and illumination mats **40** (“dual-mode mats” or “mats”) upon which items are placed. In some embodiments, the venue **100** includes multiple central controllers **16** and/or switches **18** connected to respective subsets of the cameras **30,31** and dual-mode mats **40** associated with the venue **100**.

[0026] Each of the cameras **30**, **31** and dual-mode mats **40** may be in wired or wireless electronic communication with the central controller **16** via the network switch **18** to support data communications therebetween. For example, in some embodiments, the cameras **30**, **31** and dual-mode mats **40** may be connected via Category 5 or 6 cables and use the Ethernet standard for wired communications. In other embodiments, the cameras **30**, **31** and dual-mode mats **40** may be connected wirelessly, using a built-in wireless transceiver, and may use the IEEE 802.11 (WiFi) and/or Bluetooth standards for wireless communications. Other embodiments may include cameras **30**, **31** and dual-mode mats **40** that use a combination of wired and wireless communication.

[0027] With simultaneous reference to FIG. 2, illustrated is a perspective view of a shelf **60** that includes a dual-mode mat **40** being monitored by the product monitoring system. More particularly, the perspective view depicts a scenario during the initial configuration of the cameras **30**, **31** when no products are placed upon the shelf **60**. Prior to initiating the configuration process, the central controller **16** may be programmed with position data (e.g., X, Y, and Z coordinates) for the shelf **60**, the cameras **30**, and/or the mat **40** with respect to the venue **100**. For example, the position data associated with the venue may indicate a range of X-Y coordinates associated with the shelf **60**, a Z coordinate for the height of each tier, and an arrangement of mats **40** on each tier. In the illustrated scenario, the position data associated with the mat **40** may indicate that the mat **40** is located on the second tier and second position of the shelf **60**.

[0028] During the configuration, the central controller **16** derives calibration data associated with the cameras **30**, **31** based on the known location of the mat **40**. For example, the calibration data may indicate a correspondence between positions within the venue **100** and the pixels of image data produced when the cameras **30,31** are configured with particular pan/tilt/roll settings.

Accordingly, during operation, when the mat **40** indicates a change in pressure data (e.g., when an object is removed from the dual-mode mat **40**), the central controller **16** may compare the current pan/tilt/roll of the cameras **30**, **31** to the calibration data to identify cameras that include the location of the mat **40** within the field of view (FOV) and determine which portions (e.g., which pixels) of the images provided by the identified cameras include image data representative of the object.

[0029] To initiate the calibration process, the central controller **16** may instruct the mat **40** to illuminate in accordance with a pre-determined illumination pattern. In the illustrated scenario, the illumination pattern is a solid illumination pattern where each light emitting diode (LED) of the mat **40** is illuminated. In other scenarios, the illumination pattern may include a bullseye pattern or other pattern that enables the central controller **16** to identify the position of an individual LED and/or a collection of LEDs of the mat **40** within the image data provided by the cameras **30, 31**. In these scenarios, the central controller **16** may derive the calibration data based on the known position the identified LED of the mat **40**. As a result, the central controller **16** can calibrate the cameras **30, 31** simultaneously and with greater accuracy than conventionally possible.

[0030] It should be appreciated that while the illustrated scenario depicts an initial calibration of the cameras **30, 31** where there are no products on the shelf **60**, in other scenarios, similar techniques may be applied after stocking the shelf **60** to re-calibrate the cameras **30, 31**. To this end, over time the camera mountings may shift slightly and/or the mechanical components that control the pan/tilt/roll of the camera may change resistivity due to wear and tear. This may cause the calibration data to become inaccurate, thereby requiring recalibration of recalibrate the cameras **30, 31**.

[0031] In some scenarios, the central controller **16** may initiate a recalibration process when there are products on the shelf **60**. In the re-calibration scenario, the central controller **16** may first determine a portion of the mat **40** that is not obscured by the products placed upon the mat **40** and generate an illumination pattern using the unobscured LEDs. As a result, the central controller **16** can recalibrate the cameras **30,31** in a manner that accounts for the dynamic conditions of the venue **100**. Additional description of the calibration process is provided in co-pending U.S. application Ser. No. _____ (attorney docket number 155282US01), the entire disclosure of which is hereby incorporated by reference.

[0032] FIG. 3A is an example schematic diagram of a dual-mode mat **140** (such as any of the mats **40** described with respect to FIG. 1 or 2) that enables the improved calibration and/or recalibration techniques described herein. As illustrated, the dual-mode mat **140** includes a surface matrix **142**. The surface matrix **142** may be a surface upon which items associated with the venue **100** are placed. Additionally, the surface matrix **142** houses a matrix that includes both a pressure sensor and an LED at each matrix position.

[0033] The dual-mode mat **140** also includes a selector **152** for selectively controlling which position in the surface matrix **142** is active. To this end, the surface matrix **142** may include a microcontroller (e.g., an Arduino microcontroller) that includes an input via which selector **152** is electrically connected to the surface mat **142**. Accordingly, the selector **152** may be a shift register configured to produce an output analog voltage signal that instructs the microcontroller of the surface matrix **142** which position is active. In response, the microcontroller of the surface matrix **142** completes an electrical path that flows from a power supply through the active position.

[0034] With simultaneous reference to FIG. 3B, illustrated is an example schematic diagram of the surface matrix **142** of the dual-mode mat **140**. The surface matrix **142** may include a plurality of row wires (e.g., R0 to R6) and a plurality of column wires (e.g., C0 to C7). As illustrated, the row wires and the column wires form intersection points **141** that represent the matrix positions of the surface matrix **142**. At each intersection point **141**, the surface matrix **142** includes a pressure sensor **151** and an LED **171** electrically coupled in parallel to the corresponding row wire and column wire.

[0035] In some embodiments, the pressure sensors **151** are high impedance pressure sensors (e.g., pressure sensors having an impedance of approximately 20,000 ohms to approximately 100,000 ohms). Accordingly, when the surface matrix **142** completes the electrical path to the intersection point **141** and runs a current in the forward-biased direction, the LED **171** will have significantly lower impedance than the pressure sensor **151**. As a result, the current will mainly flow through the LED **171** causing the LED **171** to illuminate. On the other hand, when the surface matrix **142**

completes the electrical path to the intersection point **141** and runs a current in the reverse-biased direction, the LED **171** acts as an open switch, causing the current to flow through the pressure sensor **151**. As a result, the pressure sensor **151** may provide an output voltage signal indicative of a pressure exerted on the mat **140**. The correspondence between the voltage signal and sensed pressure is typically noted in the product specifications associated with the pressure sensors **151**. [0036] During operation, a controller **145** controls the dual-mode mat **140** to operate in either (i) a pressure-sensing mode during which the controller **145** obtains pressure data from the pressure sensors **151**, or (ii) an illumination mode during which the LEDs **171** are illuminated in accordance with an illumination pattern. Starting with the pressure-sensing mode, the controller **145** may be configured to control the selector **152** in tandem with a switch bank **154** to scan the outputs of each of the pressure sensors **151** within the surface matrix **142**. For example, the controller **145** may maintain a matrix of sensed pressure values corresponding to each of the pressure sensors **151**. Accordingly, the controller **145** may configure the selector **152** and the switch bank **154** to complete an electrical path from a power source of the surface matrix **142** through a pre-determined pressure sensor **151** and the switch bank **154** and to an input port of the controller **145**. It should be appreciated that the surface matrix **142** may configure the current in the completed circuit to flow in a reverse-biased direction with respect to the LEDs **171** thereby causing the current to flow through the pressure sensors **151**. Based on the sensed voltage levels at the input port, the controller **145** may update the pressure values for the scanned pressure sensors **151** in the maintained matrix of pressure values.

[0037] When updating the pressure values, the controller **145** may detect a change in pressure that exceeds a threshold value for alerting a central controller (such as the central controller **16** of FIG. **1** or **2**). In one example, the threshold is approximately 80% of the maximum resistance of a variable resistor in the pressure sensors **151**. For a 3.3V system with pressure sensors that have a maximum resistance of approximately 100,000 ohms, the voltage threshold for detecting a pressure change at the A/D input port of the controller **145** is about 0.66V. If the pressure value reported by a pressure sensor **151** decreased by a threshold amount, then an item was likely removed from the mat **140**. On the other hand, if the pressure value reported by a pressure sensor **151** increased by a threshold amount, then an item was likely placed onto the mat **140**. In some embodiments, the controller **145** may generate and transmit an event to the central controller **16** that includes an identifier of the mat **140** and/or the particular pressure sensor(s) **151** that exhibited the change in pressure values. The central controller **16** may then query a planogram of the venue **100** to identify a physical location of the mat **140** and/or the identified pressure sensor **151** to obtain image data of the physical location captured by the cameras **30**, **31**.

[0038] Turning to the illumination mode, the controller **145** may be configured to control switches **173** and/or resistor bank **174** to adjust operation of the LEDs **171** of the surface matrix **142** in accordance with an illumination pattern. In some embodiments, the illumination pattern indicates a simple binary or Boolean flag as to whether each of the LEDs **171** should be enabled or disabled. In these embodiments, the controller **145** may instruct the selector **152** to cyclically select the intersection points **141** corresponding the enabled LEDs **171** in the illumination pattern. The microcontroller, in turn, completes an electrical path that flows from a power supply through the intersection point **141** to drive the enabled LEDs **171**. It should be appreciated that in the illumination mode, the current flows through the surface matrix **142** in a forward-biased direction with respect to the LEDs **171** thereby by causing the LEDs **171** to illuminate.

[0039] In other embodiments, the illumination pattern may include more complex illumination features. As one example, the illumination pattern may indicate an illumination intensity for the LEDs **171**. As another example, the LEDs **171** may be multi-color, voltage-variable LEDs that are configurable to generate different colored illumination based on the input voltage. Accordingly, in these embodiments, the illumination pattern may include a color value (e.g., a vector of red, green, and blue values) and/or an intensity value (e.g., a value from 0 to 255).

[0040] In these embodiments, a shift register (e.g., the selector **152**) may be unable to sufficiently control the LEDs **171**. Instead, the mat **140** may include an encoder **172** to encode the intensity and/or color settings of the illumination pattern into control signals for controlling the microcontroller of the surface matrix **142**. For example, the encoder **172** may be a TI 74HC4051 or a custom ASIC or FPGA encoder that preferably includes a tri-state output. Accordingly, in these embodiments, the mat **140** may include an OR gate **143** configured to receive the control signals from either the selector **152** or the encoder **172**. While the instant disclosure uses a different name to refer to the encoder **172**, in these embodiments, the encoder **172** may nonetheless be considered a “selector.”

[0041] Additionally, the mat **140** includes a resistor bank **174** controlled in tandem with the selector **152** and/or the encoder **172** to control the illumination provided by the enabled LEDs **171**. In embodiments that support simple illumination patterns, the controller **145** may set switches in a switch bank **173** to control an overall amount of resistance exhibited by the resistor bank **174** in the circuit between the selector **152**, the surface matrix **142** and the resistor bank **174**. In particular, the controller **145** may control the switch bank **173** such that the resistance of the resistor bank **174** limits the current of the circuit to be within an operating range of the LEDs **171** and/or below a pre-determined maximum current level.

[0042] In embodiments with more complex illumination patterns that indicate an illumination intensity, the resistor bank **174** may include variable gain resistors instead of fixed-gain resistors. Accordingly, the controller **145** may set the resistance of the variable-gain resistors to adjust the current provided to the LEDs **171** to effect the indicated illumination intensity. In some embodiments with illumination patterns that indicate an illumination color, the resistor bank **174** may include a resistor for each of the red, blue, and green channels. Additionally or alternatively, the resistor bank **174** may include one or more controllable switches (e.g., MOSFET transistors) for modulating the resistance in accordance with the color settings. That is, the resistor bank **174** may implement pulse-width modulation (PWM) techniques with a duty cycle configured to change the effective voltage supplied to the LEDs **171**. Because the LEDs **171** may be multi-color, voltage variable LEDs, the change in the effective voltage causes the LEDs **171** to change their illumination color. As a result, the controller **145** may control the PWM duty cycle of the resistor bank to implement the color settings of the illumination pattern.

[0043] As illustrated, the mat **140** also includes a switch **146**. The controller **145** may operate the switch **146** to connect the surface matrix **142** to the switch bank **154** when operating in the pressure-sensing mode and to connect the surface matrix **142** to the resistor bank **174** when operating the illumination mode.

[0044] Turning to FIG. 3C, illustrated is a side view of the dual-mode mat **140**, as disclosed herein. More particularly, illustrated is a side view of the surface matrix **142** of the dual-mode mat **140**.

[0045] As illustrated the surface matrix **142** includes a light permeable layer upon **144** upon which items are placed. The light-permeable nature of the layer **144** enables the illumination provided by the LEDs **171** to pass through the layer **144** and be detectable by the cameras **30**, **31** of the venue **100**. Additionally, the illustrated surface matrix **142** includes a printed circuit board (PCB) **148** upon which the pressure sensors **151** and the LEDs **171** are mounted. The PCB **148** may also include the plurality of row wires and column wires that connect to the pressure sensors **151** and the LEDs **171**.

[0046] As illustrated, the height of the LEDs **171** is less than the height of the pressure sensors **151** such that the light permeable layer **144** rests upon the pressure sensors **151**, and not the LEDs **171**. To this end, if the LEDs **171** were to have approximately the same height as the pressure sensors **151**, the LEDs **171** would support some of the pressure applied by items placed upon the surface **144** resulting in inaccurate pressure measurements by the pressure sensors **151**.

[0047] FIG. 4A illustrates an example view of the dual-mode mat **140** being configured to illuminate in accordance with an illumination pattern **178**, as disclosed herein. As illustrated, the

illumination pattern **178** may form three rectangles: an upper left rectangle, an upper right rectangle, and a lower rectangle.

[0048] With simultaneous reference to FIG. **4B**, illustrated is an example data structure for the illumination pattern **178**. As illustrated, the data structure for the illumination pattern **178** may be a matrix of values that correspond to the LEDs **171** of the surface matrix **142** in a 1:1 relationship. For example, matrix position (0,0) of the data structure may correspond to the LED **171.sub.0,0**. Accordingly, to cause the LEDs **171** to illuminate in accordance with the illumination pattern **178**, the controller **145** may analyze the illumination pattern **178** to identify the illumination settings for each of the LEDs **171** and configure the selection **152** and/or the encoder **172** to control the surface matrix **142** in accordance therewith. More particularly, the controller **145** may configure the resistor bank **174** and the switches **173** in accordance with the illumination settings indicated by the illumination pattern **178**.

[0049] In the illustrated example, the illumination pattern **178** represents a single-color illumination pattern. Accordingly, each matrix position only corresponds to a single value—illumination intensity. In alternate embodiments that include multi-color illumination patterns, each matrix position may include a vector of values that correspond to a red intensity value, a green intensity value, and a blue intensity value to use when configuring the resistance of the red, green, and blue channel resistors of the resistor bank **174**. In still other embodiments, the matrix position may include a binary or Boolean variable indicating whether the LED **171** at the corresponding position **141** is to be enabled or not.

[0050] As disclosed herein, the controller **145** may store the illumination pattern **178** in a memory. For example, the controller **145** may have received the illumination pattern **178** from the central controller **16**. In some embodiments, the controller **145** stores multiple illumination patterns in the memory. To this end, different illumination patterns may be designed for different purposes. For example, a first illumination pattern may be designed to facilitate a camera calibration process and a second illumination pattern may be designed to facilitate an assisted restocking application. Additionally, in some embodiments, the controller **145** may dynamically generate and store an illumination pattern based upon pressure data sensed during the pressure-sensing mode.

[0051] FIG. **5A** illustrates an example schematic diagram of an alternate embodiment for a dual-mode mat **240**, as disclosed herein. With simultaneous reference to FIG. **5B**, illustrated is an example diagram of a surface matrix **242** included within the alternate embodiment of the dual-mode mat **240**.

[0052] As illustrated, the LEDs **271** at each intersection point **241** in the surface matrix **242** are coupled in parallel to the pressure sensors **251** and biased in the opposite direction as the LEDs **171** of the sensor matrix **142**. As such, the direction of current flowing through the sensor matrix **242** has the opposite effect as the sensor matrix **142**. That is, the forward-biased direction of current in the surface matrix **142** is the reverse-biased direction of current in the surface matrix **242**. Similarly, the reverse-biased direction of current in the surface matrix **142** is the forward-biased direction of current in the surface matrix **242**. As a result, the current flow for the illumination mode flows in the opposite directions for the dual-mode mat **240**.

[0053] Accordingly, as shown in FIG. **5A**, the illumination mode circuitry is arranged in the opposite direction as the corresponding components of the dual-mode mat **140** to maintain their circuit position with respect to the forward direction of bias of the LEDs **271**. That is, the selector **272** for the illumination mode in the mat **240** is positioned to the left of surface matrix **242**; whereas the selector **152** for the illumination mode in the mat **140** is positioned above the surface matrix **142**. Similarly, the resistor bank **274** and the corresponding switch bank **273** of the mat **240** are positioned on the upper side of the surface matrix **242** whereas the resistor bank **174** and switch bank **173** of the mat **140** are positioned to the left of the surface matrix **142**. It should be appreciated that while the dual-mode mat **140** may use the selector **152** for both the pressure-sensing mode and the illumination mode, the dual mode mat **240** includes the selector **272** (and/or

encoder) for the illumination mode in addition to the selector **252** for the pressure-sensing mode. [0054] It should be appreciated that despite the illumination mode components of the dual-mode mat **240** being positioned in a different location to account for the different direction of bias of the LEDs **271**, the components of the dual-mode mat **240** may generally function in an equivalent manner as their corresponding component of the dual-mode mat **140**. That said, because the dual-mode mat **240** includes different selectors for the illumination mode and the pressure-sensing mode, the dual-mode mat **240** includes switches **246a-b** on both sides of the surface matrix **242** such that the controller **245** operates in tandem when switching between the pressure-sensing mode and the illumination mode. It should be appreciated that the switches **256a-b** may include multiple physical switches such that the switches **256a-b** can process tri-state inputs to the surface matrix **242**.

[0055] FIG. **6** is a block diagram representative of a controller **645** of a dual-mode mat (such as the dual-mode mats **40**, **140**, **240**) and the various components thereof. The controller **645** is represented as a logic platform that functions as a processing platform capable of executing instructions to, for example, implement functionality of the example systems described herein (including functionality described with respect to the controllers **145**, **245**). Example processing platforms include, for example, implementing operations of the example methods described herein via field programmable gate arrays (FPGAs) and application specific integrated circuits (ASICs). [0056] The controller **645** includes a processor **602** that includes a processor such as, for example, one or more microprocessors, controllers, and/or any suitable type of processor. The controller **645** includes memory (e.g., volatile memory, non-volatile memory) **604** accessible by the processor **602** (e.g., via a memory controller). The example processor **602** interacts with the memory **604** to obtain, for example, machine-readable instructions stored in the memory **604** corresponding to, for example, the controller functionality described throughout this disclosure. Additionally or alternatively, machine-readable instructions corresponding to the example operations described herein may be stored on one or more removable media (e.g., a compact disc, a digital versatile disc, removable flash memory, etc.) that may be coupled to the controller **645** to provide access to the machine-readable instructions stored thereon. In some embodiments, the memory **604** maintains one or more data structures representative of illumination patterns and/or a matrix of current pressure values sensed by the pressure sensors of a surface matrix (such as the surface matrices **142**, **242**).

[0057] The controller **645** also includes a network interface **606** to enable communication with other machines (e.g., the central controller **16** and/or other dual-mode mats) via, for example, one or more networks. The example network interface **606** includes any suitable type of communication interface(s) (e.g., wired and/or wireless interfaces) configured to operate in accordance with any suitable protocol(s). For example, the network interface **606** may include a first set of transceivers for communications with the central controller **16** and a second set of transceivers for communicating with other dual-mode mats.

[0058] In some embodiments, the controller **645** may transmit pressure data obtained during the pressure-sensing mode to the central controller **16**. For example, as described above, during the pressure-sensing mode, the controller **645** may scan the pressure sensors of a surface matrix. If the pressure value reported by any of the pressure sensors has changed by a threshold amount from the value maintained in the memory **604**, then the controller **645** may generate a pressure change event to transmit to the central controller **16**. The pressure change event may indicate a time stamp associated with the pressure data, an identifier of the pressure sensor (e.g., an identifier that indicates a position of the pressure sensor within the surface matrix), an identifier of the controller **645**, and/or whether the change in pressure is an increase or a decrease in pressure.

[0059] Additionally, the controller **645** may be configured to receive illumination patterns from the central controller **16**. For example, the central controller **16** may transmit an illumination pattern to the controller **645** for storage in the memory **604**. When the central controller **16** instructs the

controller **645** to operate in the illumination mode, the controller **645** may obtain the stored illumination pattern and control the LEDs of the surface matrix to illuminate in accordance therewith.

[0060] The controller **645** also includes I/O interfaces **608** to enable receipt of user input and communication of output data to the user.

Embodiments of a Dual-Mode Mat

[0061] Embodiments of a dual-mode mat (such as the dual-mode mats **40**, **140**, **240**) disclosed herein include a surface (such as the surface matrices **142**, **242**) that includes a plurality of row wires and a plurality of column wires forming intersection points (such as intersection points **141**, **241**). The intersection points between row wires and column wires include a pressure sensor (such as a pressure sensor **151**, **251**) (i) coupled to the row wire and the column wire at the intersection point and (ii) configured to sense a pressure exerted upon the surface, and a light emitting diode (LED) (such as the LEDs **171**, **271**) coupled to the row wire and the column wire at the intersection point in parallel to the pressure sensor. In some embodiments, the pressure sensors have an impedance from about 20,000 ohms to about 100,000 ohms.

[0062] In some embodiments, the surface includes a first, light-permeable layer (such as the layer **144**) upon which objects are placed and a second layer (such as the layer **148**) that includes a printed circuit board to which the pressure sensors and LEDs are coupled.

[0063] The dual-mode mat also includes a selector (such as the selectors **152**, **252**, **272** and the encoder **172**) configured to selectively complete an electrical path that flows from a power supply through a selected intersection point. For example, the selector may be a shift register. It should be appreciated in some embodiments, such as the embodiment depicted in FIG. 5A, the selector may include multiple component selectors.

[0064] The dual-mode mat also includes a controller (such as the controllers **145**, **245**, **645**) coupled to the selector and configured to control operation of the surface. The controller may be configured to operate in one of a pressure-sensing mode or an illumination mode.

[0065] The dual-mode mat also includes a switch bank (such as the switch banks **154**, **254**) coupled to the controller and the surface, wherein the controller is configured to operate the switch bank in tandem with the selector when operating in the pressure-sensing mode.

[0066] The dual-mode mat also includes a resistor bank (such as the resistor banks **174**, **274**) coupled to the controller and the surface, wherein the controller is configured to operate the resistor bank in tandem with the selector when operating in the illumination mode.

[0067] The dual-mode mat also includes a switch (such as the switches **146**, **246a**, **246b**) operatively coupled to the controller, wherein (i) when the controller is configured to operate in the pressure-sensing mode, the controller operates the switch such that the switch completes a circuit between the selector, the surface, and the switch bank, and (ii) when the controller is configured to operate in the illumination mode, the controller operates the switch such that the switch completes a circuit between the selector, the surface, and the resistor bank. It should be appreciated that in some embodiments, such as the embodiment depicted in FIG. 5A, the switch may include multiple component switches.

[0068] In some embodiments, to control operation of the surface in the illumination mode, the controller is configured to configure current flowing through the electrical path to the selected intersection point to flow in a first direction such that the LED is forward-biased. Similarly, to control operation of the surface in the pressure-sensing mode, the controller is configured to configure current flowing through the electrical path to the selected intersection point to flow in a first direction such that the LED is reverse-biased. It should be appreciated that the first and second directions may vary depending upon the orientation of the LEDs of the surface. For example, the “first direction” of the surface **142** is a different direction as the “first direction” of the surface **242**.

[0069] In some embodiments, to control operation of the surface in the illumination mode, the controller is configured to obtain, from a memory (such as the memory **604**) of the controller, an

illumination pattern that indicates one or more intersection points at which the LED is to be illuminated. The controller may then control the selector in accordance with the illumination pattern by causing the selector to cyclically select the indicated intersection points.

[0070] In some embodiments, the illumination pattern indicates an illumination intensity for the one or more illuminated intersection points. In these embodiments, the controller configures operation of the resistor bank in accordance with the indicated illumination intensities such that when the illuminated intersection points are selected by the selector, the LEDs of the illuminated intersection points are illuminated based on the indicated intensities. It should be appreciated that the indicated intensity may not represent an actual intensity unit of measurement (e.g., candelas) but a factor by which a maximum intensity of the LED is scaled. For example, if the illumination intensity values use a scale from 0 to 255, the illumination intensity may indicate a numerator of a fraction having 256 as the denominator. The controller may then scale the illumination intensity of the LED based on the resulting fraction.

[0071] In some embodiments where the LEDs are multi-color, voltage-variable LEDs, the illumination pattern indicates a color for the one or more illuminated intersection points. In these embodiments, the controller configures operation of the resistor bank in accordance with the indicated illumination colors such that when the illuminated intersection points are selected by the selector, the LEDs of the illuminated intersection points are illuminated in the indicated colors.

[0072] In some embodiments, to control operation of the surface in the pressure-sensing mode, the controller is configured to (i) control the selector to cyclically select the intersection points; (ii) configure the switch bank to complete an electrical path from the pressure sensor to an input terminal of the controller; and (iii) obtain a pressure value at the selected intersection point based upon a voltage level detected at the input terminal of the controller.

[0073] In some embodiments, the controller additionally comprises one or more transceivers (such as the network interface **606**) coupled to one or more remote computing systems. In these embodiments, the one or more transceivers may be coupled to a central controller (such as the central controller **16**). For example, the controller may be configured to receive illumination patterns (such as the illumination pattern **178**) from central controller, and transmit indications of sensed pressure values to the central controller.

Modular Arrangement of Dual-Mode Mats

[0074] As shown in FIG. 2, the shelf **60** may include multiple dual-mode mats **40** on a tier. In the illustrated example, the second tier includes four dual-mode mats **40**. In many embodiments, the dual-mode mats **40** communicate with the central server **16** via an IEEE 802.11 (Wi-Fi) communication link. For each of the dual-mode mats **40** to directly communicate with the central controller **16** in the above-described manner, then each of the dual-mode mats **40** would need to include the corresponding Wi-Fi communication components, adding additional cost and size to each dual-mode mat **40**.

[0075] Accordingly, techniques described herein relate to connecting multiple dual-mode mats in a modular arrangement of supervisor nodes and subordinate nodes. In these arrangements, only the supervisor nodes may include the Wi-Fi circuitry for communication with the central controller **16**. As a result, the more numerous subordinate nodes may be constructed at a lower cost. Additionally, the subordinate nodes may have a smaller form factor for the control components thereby enabling additional items to be placed on the surfaces thereof.

[0076] FIG. 7 illustrates a modular arrangement **700** of dual-mode mats **740** coupled via a communication bus **748**, in accordance with techniques described herein. For example, the four mats **740** may correspond to the four mats on the second tier of the shelf **60** of FIG. 2.

[0077] As illustrated, the mats **740** include a supervisor node **740a** and three subordinate nodes **740b-d**. Each of the dual-mode mats **740** may be configured in a similar manner as the dual-mode mats **40**, **140**, and **240**. However, unlike the subordinate nodes **740b-d**, the controller **745a** of the supervisor node **740a** includes a wireless transceiver **747** for communication with a central

controller **716** (such as the central controller **16**).

[0078] Instead, the controllers **745b-d** of the subordinate nodes are configured to communicate over a communication bus **748**. The communication bus **748** may implement any suitable bus communication protocol, including proprietary bus communication protocols. Additionally, the controller **745a** of the supervisor node **740a** is also coupled to the communication bus **748**. Accordingly, the supervisor node **740a** may act as a relay between the central controller **716** and the subordinate nodes **740b-d**.

[0079] As one example, the central controller **716** may transmit an illumination pattern for the subordinate node **740c** to the supervisor node **740a** over a Wi-Fi network. In this example, the supervisor node **740a** may then transmit an instruction over the communication bus **748** instructing the subordinate node **740c** to store the received illumination pattern in a memory thereat. In some embodiments, when the subordinate node **740c** receives the illumination pattern over the communication bus **748** the controller **745c** of the subordinate node **740c** may begin operating in the illumination mode. In other embodiments, the subordinate node **740c** may simply store the illumination pattern in the memory and await further instruction before operating in the illumination mode.

[0080] As another example, when operating in the pressure-sensing mode, the controllers **745b-d** of the subordinate nodes **740b-d** may continuously scan their respective pressure sensors to determine if a change in pressure value exceeds a threshold amount. Accordingly, when a controller **745b-d** detects that a pressure value changed by a threshold amount, the controller **745b-d** may generate and transmit a pressure change event over the communication bus **748**. The pressure change event may indicate a time stamp associated with the pressure data, an identifier of the pressure sensor (e.g., an identifier that indicates a position of the pressure sensor within the surface matrix), an identifier of the controller **745b-d**, and/or whether the change in pressure is an increase or a decrease in pressure. As such, the pressure change event may be considered an indication of sensed pressure values. The controller **745a** of the supervisor node **740a** may detect that a pressure change event was written to the communication bus **748** and transmit the event to the central controller **716** via the Wi-Fi network.

[0081] Additionally, each of the dual-mode mats **740a-d** may share a common power source **790**. In some embodiments, the power source **790** includes an energy harvester for generating the power. For example, the power source **790** may include a solar, thermal, radio frequency (RF), and/or piezoelectric energy harvester. In addition to the energy harvester, the power source **790** may include a fallback DC power source.

[0082] FIG. **8** illustrates an alternate modular arrangement of dual-mode mats **840** coupled via a daisy chain arrangement, in accordance with techniques described herein. The dual-mode mats **840a-c** may be configured in a similar manner as the dual-mode mats **740a-c**. That is, the supervisor node **840a** acts as a relay between the central controller **816** and the subordinate node **840b-c**.

[0083] However, rather than relaying the communications over the communication bus **748**, the modular arrangement **800** includes a daisy chain of direct communicative couplings between the dual-mode mats **840**. That is, the supervisor node **840a** is coupled to the subordinate node **840b** via the direct coupling **848a** and the subordinate node **840b** is coupled to the subordinate node **840c** via the direct coupling **848b**. Accordingly, if the controller of the subordinate node **840c** detects a change in pressure, the controller sends the pressure change event to the subordinate node **840b** via the direct coupling **848b**, which relays the pressure change event to the supervisor node **840a** via the direct coupling **848a**, which relays the pressure change event to the central controller **816** via the Wi-Fi connection facilitated by the Wi-Fi transceiver **847**. Similarly, if the central controller **816** wants to send an illumination pattern to the subordinate node **840c**, the central controller **816** sends the illumination pattern to the supervisor node **840a** via the Wi-Fi connection facilitated by the Wi-Fi transceiver **847**, which relays the illumination pattern to the subordinate node **840b** via

the direct coupling **848a**, which relays the illumination pattern to the subordinate node **840c** via the direct coupling **848b**.

[0084] The above description refers to a block diagram of the accompanying drawings. Alternative implementations of the example represented by the block diagram includes one or more additional or alternative elements, processes and/or devices. Additionally or alternatively, one or more of the example blocks of the diagram may be combined, divided, re-arranged or omitted. Components represented by the blocks of the diagram are implemented by hardware, software, firmware, and/or any combination of hardware, software and/or firmware. In some examples, at least one of the components represented by the blocks is implemented by a logic circuit. As used herein, the term “logic circuit” is expressly defined as a physical device including at least one hardware component configured (e.g., via operation in accordance with a predetermined configuration and/or via execution of stored machine-readable instructions) to control one or more machines and/or perform operations of one or more machines. Examples of a logic circuit include one or more processors, one or more coprocessors, one or more microprocessors, one or more controllers, one or more digital signal processors (DSPs), one or more application specific integrated circuits (ASICs), one or more field programmable gate arrays (FPGAs), one or more microcontroller units (MCUs), one or more hardware accelerators, one or more special-purpose computer chips, and one or more system-on-a-chip (SoC) devices. Some example logic circuits, such as ASICs or FPGAs, are specifically configured hardware for performing operations (e.g., one or more of the operations described herein and represented by the flowcharts of this disclosure, if such are present). Some example logic circuits are hardware that executes machine-readable instructions to perform operations (e.g., one or more of the operations described herein and represented by the flowcharts of this disclosure, if such are present). Some example logic circuits include a combination of specifically configured hardware and hardware that executes machine-readable instructions.

[0085] As used herein, each of the terms “tangible machine-readable medium,” “non-transitory machine-readable medium” and “machine-readable storage device” is expressly defined as a storage medium (e.g., a platter of a hard disk drive, a digital versatile disc, a compact disc, flash memory, read-only memory, random-access memory, etc.) on which machine-readable instructions (e.g., program code in the form of, for example, software and/or firmware) are stored for any suitable duration of time (e.g., permanently, for an extended period of time (e.g., while a program associated with the machine-readable instructions is executing), and/or a short period of time (e.g., while the machine-readable instructions are cached and/or during a buffering process)). Further, as used herein, each of the terms “tangible machine-readable medium,” “non-transitory machine-readable medium” and “machine-readable storage device” is expressly defined to exclude propagating signals. That is, as used in any claim of this patent, none of the terms “tangible machine-readable medium,” “non-transitory machine-readable medium,” and “machine-readable storage device” can be read to be implemented by a propagating signal.

[0086] In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings. Additionally, the described embodiments/examples/implementations should not be interpreted as mutually exclusive, and should instead be understood as potentially combinable if such combinations are permissive in any way. In other words, any feature disclosed in any of the aforementioned embodiments/examples/implementations may be included in any of the other aforementioned embodiments/examples/implementations.

[0087] The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The claimed invention is

defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

[0088] Moreover, in this document, relational terms such as first and second, top and bottom, and the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has”, “having,” “includes”, “including,” “contains”, “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

[0089] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter may lie in less than all features of a single disclosed embodiment. Thus, the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. An apparatus comprising: a surface that includes a plurality of row wires and a plurality of column wires forming intersection points, wherein intersection points between row wires and column wires include: a pressure sensor (i) coupled to the row wire and the column wire at the intersection point and (ii) configured to sense a pressure exerted upon the surface, and a light emitting diode (LED) coupled to the row wire and the column wire at the intersection point in parallel to the pressure sensor; a selector configured to selectively complete an electrical path the flows from a power supply through a selected intersection point; and a controller coupled to the selector and configured to control operation of the surface, wherein the controller is configured to operate in one of a pressure-sensing mode or an illumination mode.
2. The apparatus of claim 1, wherein the intersection points between row wires and column wires further include: a switch bank coupled to the controller and the surface, wherein the controller is configured to operate the switch bank in tandem with the selector when operating in the pressure-sensing mode; a resistor bank coupled to the controller and the surface, wherein the controller is configured to operate the resistor bank in tandem with the selector when operating in the illumination mode; and a switch operatively coupled to the controller, wherein: when the controller is configured to operate in the pressure-sensing mode, the controller operates the switch such that

the switch completes a circuit between the selector, the surface, and the switch bank, and when the controller is configured to operate in the illumination mode, the controller operates the switch such that the switch completes a circuit between the selector, the surface, and the resistor bank.

3. The apparatus of claim 1, wherein the surface includes a first, light-permeable layer upon which objects are placed and a second layer that includes a printed circuit board to which the pressure sensors and LEDs are coupled.

4. The apparatus of claim 1, wherein the pressure sensor has an impedance from about 20,000 ohms to about 100,000 ohms.

5. The apparatus of claim 2, wherein to control operation of the surface in the illumination mode, the controller is configured to: configure current flowing through the electrical path to the selected intersection point to flow in a first direction such that the LED is forward-biased.

6. The apparatus of claim 5, wherein to control operation of the surface in the illumination mode, the controller is configured to obtain, from a memory of the controller, an illumination pattern that indicates one or more intersection points at which the LED is to be illuminated; and control the selector in accordance with the illumination pattern by causing the selector to cyclically select the indicated intersection points.

7. The apparatus of claim 6, wherein: the illumination pattern indicates an illumination intensity for the one or more illuminated intersection points, and to control operation of the surface in the illumination mode, the controller is configured to: configure operation of the resistor bank in accordance with the indicated illumination intensities such that when the illuminated intersection points are selected by the selector, the LEDs of the illuminated intersection points are illuminated at the indicated intensities.

8. The apparatus of claim 6, wherein: the LEDs are multi-color, voltage-variable LEDs; the illumination pattern indicates a color for the one or more illuminated intersection points, and to control operation of the surface in the illumination mode, the controller is configured to: configure operation of the resistor bank in accordance with the indicated illumination colors such that when the illuminated intersection points are selected by the selector, the LEDs of the illuminated intersection points are illuminated in the indicated colors.

9. The apparatus of claim 2, wherein to control operation of the surface in the pressure-sensing mode, the controller is configured to: configure current flowing through the electrical path to the selected intersection point to flow in a first direction such that the LED is reverse-biased.

10. The apparatus of claim 9, wherein to control operation of the surface in the pressure-sensing mode, the controller is configured to control the selector to cyclically select the intersection points; configure the switch bank to complete an electrical path from the pressure sensor to an input terminal of the controller; and obtain a pressure value at the selected intersection point based upon a voltage level detected at the input terminal of the controller.

11. The apparatus of claim 1, wherein the controller comprises: one or more transceivers coupled to one or more remote computing systems.

12. The apparatus of claim 11, wherein: the one or more transceivers are coupled with a central controller; and the controller is configured to: receive illumination patterns from central controller, and transmit indications of sensed pressure values to the central controller.

13. A modular apparatus comprising: a first apparatus of claim 1 configured to operate as a supervisor apparatus; and a second apparatus of claim 1 configured to operate as a subordinate apparatus; wherein: the controller of the first apparatus is coupled to the controller of the second apparatus, and the controller of the first apparatus is configured to: transmit illumination patterns to the second apparatus, and receive indications of sensed pressure values from the second apparatus.

14. The modular apparatus of claim 13, further comprising a third apparatus of claim 10 configured to operate as a subordinate apparatus.

15. The modular apparatus of claim 14, further comprising: a communication bus via which the controller of the first apparatus, the controller of the second apparatus, and the controller of the

third apparatus are communicatively coupled.

16. The modular apparatus of claim 15, wherein the controller of the first apparatus is configured to: transmit illumination patterns to the second apparatus and the third apparatus via the communication bus, and receive indications of sensed pressure values from the second apparatus and the third illuminated apparatus via the communication bus.

17. The modular apparatus of claim 14, wherein: the controller of the second illuminated apparatus is configured to: transmit illumination patterns to the third apparatus, and receive indications of sensed pressure values from the third apparatus.

18. The modular apparatus of claim 17, wherein: the controller of the second apparatus is configured to: relay (i) illumination patterns received from the first apparatus to the third apparatus, and (ii) indications of sensed pressure values received from the third apparatus to the first apparatus.

19. The modular apparatus of claim 13, wherein the controller is: communicatively coupled with a central controller; and configured to: receive illumination patterns from central controller, and transmit indications of sensed pressure values to the central controller.

20. The modular apparatus of claim 19, wherein the received illumination patterns include the illumination patterns transmitted to the second apparatus.

21. The modular apparatus of claim 19, wherein the transmitted indications of sensed pressure values include the indications of sensed pressure values received from the second apparatus.
